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Predictors of Improvement in Subjective Executive Functioning Following an Internet-Delivered Cognitive Enhancement Intervention for Adults in Remission From Depression

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ABSTRACT

Introduction: Residual cognitive deficits are commonly reported by individuals in remission from depression, often affecting daily life functioning and mental health. To provide tailored and personalized cognitive enhancement interventions for this population, there is a need for a better understanding of the characteristics of those who benefit from such interventions. Therefore, this study aimed to identify predictors of changes in subjective executive functioning following an internet-delivered cognitive enhancement intervention for adults in remission from depression.

Methods: Data were collected from a randomized controlled trial investigating the efficacy of an internet-delivered cognitive enhancement intervention. Changes in subjective executive functioning from pre-treatment to the six-month follow-up were assessed in 44 participants in remission from depression, using the Behavior Rating Inventory of Executive Function Adult Global Executive Composite. Linear mixed model analyses were conducted to investigate the impact of demographic, clinical, and treatment credibility variables on change in subjective cognitive functioning over time.

Results: The results showed that shorter lifetime depression duration predicted greater improvements in subjective executive functioning ($p = 0.031$). Higher levels of treatment expectancy and credibility were related to greater improvements in subjective cognitive functioning ($p = 0.024$). Participants with a partner showed better treatment response than those without a partner ($p < 0.001$).

Conclusion: This study builds on previous research on cognitive enhancement interventions in remitted depression, highlighting the impact of depression duration, treatment expectancy, and credibility on treatment response. Interventions targeting cognitive deficits appear most effective for those with a shorter lifetime duration of depression. Therefore, efforts should be made to enhance outcomes in those with a chronic course. To maximize engagement and outcomes, these interventions should be delivered in a way that individuals in remission from depression view them as credible and capable of reducing their deficits.

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Previous research has not found partner status to predict change in subjective executive functioning. The effect of partner status on treatment response should be investigated further.

1 | Introduction

Cognitive deficits are common during remission from depression, where small to large deficits are observed in performance on neuropsychological tests measuring aspects of different cognitive domains [1]. Most cognitive enhancement interventions have targeted objective cognitive deficits measured using neuropsychological tests [2]. However, subjective experiences of cognitive deficits are also important to investigate, as more than 40% of formerly depressed adults in remission self-report such challenges [3]. Subjective cognitive deficits also negatively impact socio-occupational performance [4], return to work after depression [5], psychosocial functioning [6, 7], quality of life [8], and increase the risk of relapse [9]. Moreover, subjective measures of cognitive functioning are suggested to capture the cognitive challenges that individuals experience in their daily life more broadly, as compared to performance in a structured test environment [10]. It is important to acknowledge, however, that subjective measures of cognitive functioning can be affected by mood symptoms [11]. Therefore, subjective cognitive functioning should be assessed during remission [12]. Overall, improving subjective cognitive functioning might be important for enhancing daily life functioning, improving quality of life, and reducing the risk of depression relapse.

Only a few randomized controlled trials have investigated the effect of cognitive enhancement interventions on subjective cognitive deficits during remission from depression. These studies have shown promising results in improving subjective cognitive functioning through digital cognitive control training (Paced Auditory Serial Addition Test; [13]), cognitive training combined with strategy guidance [14], and internet-delivered cognitive enhancement therapy including psychoeducation, attention training, and compensatory strategies [15]. However, it is likely that some individuals benefit more than others from cognitive interventions, given the heterogeneity in clinical symptoms, domains of cognitive impairment, and comorbidity across individuals with a history of depression [16, 17]. Therefore, future cognitive enhancement interventions should address this variability by providing adaptive and personalized interventions, as one standard intervention is not likely to fit all [2]. Personalization may involve tailoring intervention content and training tasks, adjusting the level of therapist guidance, and modifying the order of content [18]. Research on internet-delivered interventions for mental health symptoms has shown that adaptive interventions are associated with lower dropout rates and improved treatment outcomes, particularly among those at risk of non-response [19]. However, the development of effective personalized and adaptive interventions for cognitive deficits following depression requires more knowledge of the predictors of treatment response.

To the knowledge of the authors, there are only two available studies that have explored predictors of treatment response to cognitive enhancement interventions for adults in remission from depression. Both studies found that the duration of depression predicted improvements in cognition [20, 21]. The open pilot trial by Myklebost et al. [21] found that individuals who had been depressed for a year or less overall during their lifetime showed greater improvement in subjective executive functioning than those with a longer lifetime duration of depression. However, individuals with a longer lifetime duration also benefited from the intervention. This open trial also found that higher levels of treatment credibility and expectancy were associated with treatment response. Listunova et al. [20] reported that the duration of depression was the only predictor of improved performance on an attention test after receiving a face-to-face cognitive enhancement intervention. No other demographic characteristics, depression symptoms, levels of self-efficacy, or rumination were identified to predict changes in cognitive functioning.

1.1 | Aims of the Study

Overall, there is limited research on predictors of change in cognitive functioning after cognitive enhancement interventions in individuals remitted from depression. The current study aims, therefore, to investigate predictors of change in subjective executive functioning after an internet-delivered cognitive enhancement intervention. It was hypothesized that: (1) Overall lifetime duration of depression will impact change in subjective executive functioning. (2) Levels of treatment expectancy and credibility will impact change in subjective executive functioning. (3) No other demographic or clinical variables will impact change in subjective executive functioning.

2 | Materials and Methods

2.1 | Design

This study is a secondary analysis of a randomized controlled cross-over trial (ClinicalTrial.gov ID: NCT04864353; [15]).

2.2 | Participants and Procedures

In the randomized crossover trial, participants were recruited from across Norway between April 2021 and October 2023 through social media and public posters. Participants were eligible if they met the following criteria: (1) Having received psychological or medical treatment for depression in primary or secondary health care; (2) Current minor depressive symptoms, defined as a score of < 12 on the MADRS-S scale [22]; (3) No diagnosis of depression within the last 12 weeks; (4) Self-reported cognitive deficits impacting daily life, assessed by a clinical psychologist; (5) Stable dosage of antidepressant medication during the intervention period; (6) Age between 18 and 65 years; and (7) Access to the internet. Participants were excluded if they: (1) Had ongoing substance abuse issues; (2) Had neurological conditions or brain damage affecting cognitive functioning (e.g., multiple sclerosis, autism, brain tumors); or (3) Had a diagnosis of bipolar disorder; (4) Had a diagnosis of psychosis.

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Summary

- Significant outcomes
  - Shorter duration of depression predicts treatment response: Consistent with previous research, individuals in remission with a history of being depressed for a year or less showed larger treatment responses than those with a longer duration of depression.
  - Impact of treatment expectancy and credibility: Higher expectations of benefits from the intervention and perceiving the intervention to be credible were related to greater improvements in subjective executive functioning.
  - Potential influence of partner status: While previous studies have not found partner status to predict treatment response, the findings from this study suggest that individuals with a partner experienced greater improvements in subjective executive functioning.
- Limitations
  - Generalizability: Sample characteristics limit the generalizability of the findings to the broader population of individuals in remission from depression.
  - Assessments: This study focused on subjective executive functioning and did not assess performance on neuropsychological tests. As subjective and objective measures of cognitive functioning often do not correlate, it is unclear which variables predict changes in objective executive functions.

A total of 44 adults in remission from depression were included in the randomized cross-over trial, where 23 were randomized to receive the internet-delivered intervention and 21 to the waitlist control group (Figure 1). Participants in the waitlist control group received the internet-delivered intervention after 7 weeks. Forty participants completed post-treatment assessment, and 37 participants responded to the six-month follow-up assessment. The current secondary analyses have pooled data from all 44 participants.

2.3 | Intervention

The intervention consisted of 10 modules delivered sequentially in a two-factor authenticated personalized online browser. Participants were encouraged to complete the intervention within 5 weeks. However, if they were unable to complete the intervention within this timeframe, they were able to continue working in the program until week nine of the intervention period. New modules were provided twice per week, depending upon whether their therapists judged the participants progression in the previous module as satisfactory. On average, participants completed 9.38 modules. Mean time from the first activity to the completion of each module was as follows: Module 1 (M = 19.86 h, SD = 27.59); Module 2 (M = 30.51 h, SD = 40.08); Module 3 (M = 15.05 h, SD = 40.26); Module 4 (M = 35.18 h, SD = 42.84); Module 5 (M = 30.51 h, SD = 51.76); Module 6 (M = 64.48 h, SD = 100.35); Module 7 (M = 28.92 h, SD = 46.76); Module 8 (M = 72.56 h, SD = 87.76); Module 9 (M = 25.06 h, SD = 49.19); Module 10 (M = 61.35 h, SD = 83.95). Number of overall logins to the intervention platform ranged from 5 to 47 (M = 22.88, SD = 10.01). The mean duration of treatment from inclusion to finishing the program was 43.08 days (ranging from 33 to 63 days). The modules targeted challenges within cognitive domains such as attention, memory, social cognition, and executive functions, as well as addressing negative beliefs and rumination. The core components of the intervention included: (1) psychoeducation on cognitive deficits and their underlying mechanisms; (2) compensatory strategies, such as the use of calendars, frequent breaks, self-talk, problem-solving, and the method of loci; (3) auditory attention training; (4) cognitive restructuring techniques. Each module provided participants with: a reflection task on how a cognitive deficits in a specific cognitive domain had impacted their daily life; vignettes of individuals with similar cognitive deficits; tasks promoting reflection on how to apply various strategies from the modules into their daily life. Participants self-tailored the intervention by selecting strategies and exercises within each module that they found most relevant to employ in their daily lives. These selected strategies were documented in an interactive table, which also included participants' evaluations of how well each strategy worked.

Participants accessed the intervention using their computer, tablet, or mobile phone. They were guided by a therapist providing weekly brief telephone calls. Therapist support was delivered by clinical psychologists or nurses. The calls were semi-structured and focused on goal-setting, goal evaluation, clarification of intervention content, and monitoring depressive symptoms, including assessment of suicidality. Therapists received training across three sessions, each lasting approximately 2 h, and participated in group-based supervision sessions every other week. These supervision sessions were led by an experienced clinical psychologist and focused on discussing challenges related to guiding participants, as well as reinforcing the study guidelines. A detailed description of the intervention can be found in [15].

2.4 | Measures

2.4.1 | Outcome Variable

The Behavior Rating Inventory of Executive Function Adult Global Executive Composite (BRIEF-A GEC; [23]) was used to measure change in subjective executive functioning from pre-assessment to six-month follow-up assessment. This 75-item questionnaire asks participants to rate the frequency of the executive difficulties they experience in daily life using a three-point scale: never, sometimes, or often. Higher scores indicate greater executive difficulties. The scores are organized into nine subscales: inhibit, shift, self-monitor, emotional control, working memory, plan/organize, task monitor, and organization of materials. These subscales can be combined to form a Global Executive Composite (GEC), which provides an overall measure of executive functioning. The BRIEF-A has demonstrated high internal consistency and has been validated in prior research [23, 24].

2.4.2 | Predictor Variables

Demographic variables included age, gender, partner status, education, and employment. Partner status and education were categorized into the following dichotomous variables: partner/no partner and higher education/no higher education.

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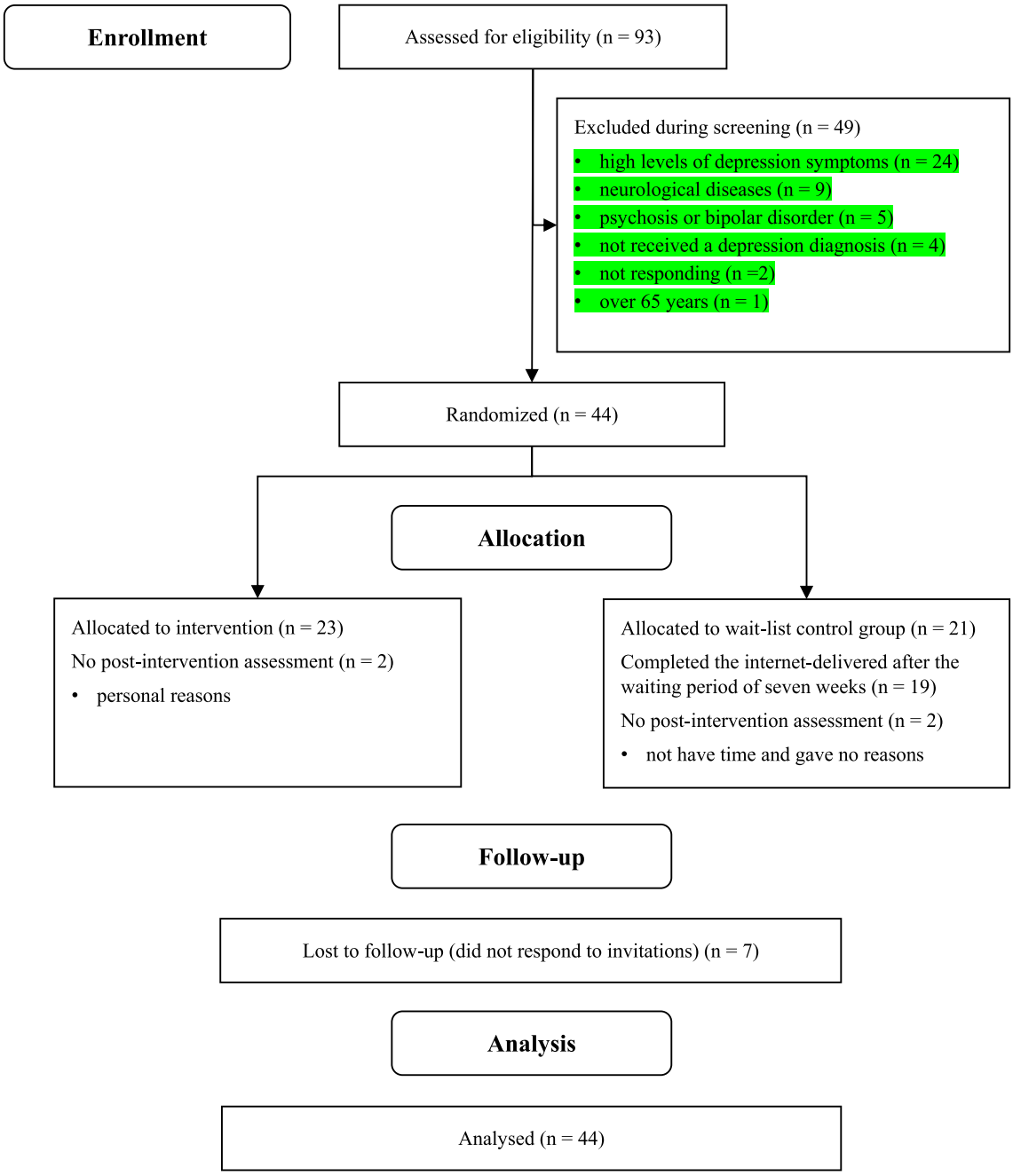


FIGURE 1 | Participant flow diagram.

Illness history variables included number of depression episodes and overall lifetime depression duration. Overall lifetime duration of depression was assessed using the following five categories: ≤ 3 months; ≤ 6 months; ≤ 1 year, ≤ 2 years, ≤ 5 years; ≤ 5 years. For the predictor analyses where these categories consolidated into a binary variable: ≤ 1 year (1 year or less) and > 1 year (more than 1 year). This binary categorization was used to distinguish individuals with chronic depression from those with non-chronic depression and to replicate the analytical approach employed in our previous study [21].

Symptom variables included rumination and depression symptoms. Rumination was measured with the Rumination Response Scale (RRS: [25]) which consists of 22 items rated from 1 (almost never) to 4 (almost always). Higher scores indicate greater levels of rumination. Symptoms of depression were measured using the Patient Health Questionnaire [26] which comprises nine items. Participants rated the severity of depression symptoms from 0 to 3.

Treatment credibility and expectancy were measured using a modified version of the Credibility and Expectancy Scale (CEQ; [27]), consisting of five items. Participants rated aspects of treatment expectancy and credibility on a scale from 0 to 10. Higher scores indicated greater levels of treatment credibility and expectancy. Participants responded to the CEQ 1 week after receiving access to the internet-delivered intervention.

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2.5 | Statistical Analyses

Statistical analyses were performed in SPSS 26. Linear mixed model analyses were used to investigate the impact of the predictor variables on the BRIEF-A GEC from pre-treatment assessment to follow-up assessment. This time interval was applied to replicate analyses in our previous predictor study [21] and to investigate predictors of long-term improvement of subjective executive functioning. Each model included two dummy-coded time variables to allow for the inclusion of all three measurement occasions (pre-treatment, post-treatment, and follow-up), with pre-treatment serving as the reference category. The first dummy variable represents the post-treatment time point (coded as: pre = 0, post = 1, follow-up = 0), and the second dummy variable represents the follow-up time point (coded as: pre = 0, post = 0, follow-up = 1). Including both dummy variables ensures that the model intercept reflects an average at the pre-treatment assessment rather than a combination of the pre-treatment and post-treatment assessments. The pre-treatment data were collected for both groups before randomization. Post-treatment assessments reflect data collected after the intervention, although the timing of these assessments varied slightly due to differences in when participants completed the intervention. Responses from the control group at the seven-week assessment were excluded from the analyses as the primary aim of this study was to investigate the changes in outcomes related to the digital intervention. Follow-up assessments were conducted 6 months after each participant's final activity in the intervention. Analyses also identified predictors of baseline values on the BRIEF-A GEC. All continuous predictor variables were mean-centered before the analyses. This implies that the intercept represents the expected outcome when all predictors are at their average value. Mean-centering enhances the interpretation of the findings, particularly for variables where 0 is not a meaningful value, such as age. The model was set as a random intercept fixed slope model. Fixed effects were estimated for time, the predictor, and the interaction between time and predictor. The estimator was fitted to restricted maximum likelihood estimation. Missing data were assumed to be missing at random. Data from all participants were included in the analyses. Statistical significance was determined at a p-value of 0.05. A Bonferroni post hoc test was applied to correct for multiple comparisons across the mixed model analyses ($p < 0.004$). P-values that remained significant after correction are highlighted in bold in the result section.

TABLE 1 | Overview of participants ($N = 44$) observed pre-treatment characteristics.

Variables	M, (SD), *n*
Age in years	37.93 (11.36)
Gender (female/male)	39/5
Education (higher/No higher)	35/9
Partner status (partner/no partner)	25/19
Depression duration (≤ 1 year/> 1 years)	15/29
Number of depressive episodes	3.02 (2.70)
BRIEF-A	132.86 (16.40)
RRS	40.68 (8.38)
PHQ-9	5.09 (2.14)
CEQ	35.88 (5.34)

Abbreviations: BRIEF-A, behavior rating inventory of executive function adult global executive composite; CEQ, credibility and expectancy scale; M, mean; PHQ-9, patient health questionnaire; RRS, rumination response scale; SD, standard deviation.

3 | Results

3.1 | Sample Characteristics

This study included 44 adults in remission from depression. The participants pre-treatment characteristics are presented in Table 1.

3.2 | Impact of Variables on Change in Cognitive Functioning

Among the continuous variables, higher levels of treatment credibility and expectancy impacted change on the BRIEF-A from pre-treatment to follow-up, while this effect did not remain significant after applying the Bonferroni correction. None of the other continuous variables (such as age, number of depression episodes, depression symptoms and rumination) predicted change. Table 2 displays the results from linear mixed model analyses for continuous variables.

TABLE 2 | Results from linear mixed model analyses with estimated pre-treatment levels and change in the BRIEF-A GEC adjusted for continuous variables.

Variable	Pre-treatment (95% CI)	*p*	Pre-treatment to FU (95% CI)	*p*
Age	−0.22 (−0.71–0.26)	0.371	0.21 (−0.37–0.80)	0.473
Episodes	0.67 (−1.40–2.74)	0.521	0.11 (−2.27–2.51)	0.922
RRS*	0.56 (−0.07–1.20)	0.082	0.16 (−0.60–0.92)	0.678
PHQ-9	1.74 (−0.85–4.32)	0.184	−0.49 (−3.58–2.69)	0.756
CEQ	0.51 (−0.54–1.56)	0.333	−1.30 (−2.42 to −0.18)	0.024*

Abbreviations: CEQ, credibility and expectancy scale; PHQ-9, patient health questionnaire; Pre-treatment to FU, estimated change in the BRIEF-A GEC per unit change in the predictor variable from pre-treatment to follow-up; Pre-treatment, estimated pre-treatment intercept levels on the BRIEF-A GEC when predictor variables are centered; RRS, rumination response scale.
*$p \leq 0.05$.

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Duration of depression was one of the categorical variables that impacted change on the BRIEF-A from pre-treatment to follow-up, with participants who had been depressed for a year or less showing greater improvements than those who had been depressed for longer periods. Having a partner was also significant, with greater improvements compared to those who did not have a partner. None of the other continuous variables, such as education and gender, predicted change. Having a partner was the only variable statistically significant after applying the Bonferroni correction. Table 3 presents the results from the linear mixed model analyses for categorical variables.

4 | Discussion

The aim of the present study was to investigate predictors of improvement in subjective executive functioning after an internet-delivered cognitive enhancement intervention among individuals remitted from depression. In line with previous studies, the findings showed that overall lifetime duration of depression and treatment credibility and expectancy predicted treatment outcome [20, 21]. However, the present study found that having a partner also predicted greater treatment response. As anticipated, the study did not find any of the other demographic variables, illness history variables, or symptom variables to impact improvement in cognitive functioning.

Consistent with the findings of previous studies, lifetime duration of depression was found to predict treatment response [20, 21]. Specifically, participants who had a history of being depressed for a year or less showed larger improvements after 6 months compared to those who had been depressed for a longer duration throughout their lifetime. This finding might be related to effort and motivation. Depressed individuals are more likely to make additional errors on cognitive tests after experiencing an initial error [28] and avoid cognitively challenging tasks [29]. Individuals who have been depressed for longer periods might have accumulated more experiences with cognitive failures in daily life [30]. Based on this, one might speculate whether experiences of cognitive challenges affect motivation and effort to engage in the tasks provided during the intervention and therefore experience less effect. Future research should investigate how effort to use tasks and strategies learned in cognitive enhancement interventions affects outcomes. Moreover, there is a need to explore approaches to improving the effects of cognitive enhancement interventions for individuals who have experienced longer durations of depression. One approach might involve providing more extensive therapist guidance and follow-up interventions. This may increase their motivation to employ skills learned in the intervention in daily life activities. The dose–response effects of cognitive enhancement interventions might vary depending on depression history, and thus establishing the right duration and frequency of cognitive enhancement interventions is of concern for future research, based on the results from this study.

In line with a previous study, the findings in the present study show that higher levels of treatment credibility and expectancy impact improvements in subjective executive functioning [21]. Several studies have shown these factors to also improve treatment outcomes of internet-delivered cognitive behavioral interventions targeting mental health disorders [31]. These are considered common factors or non-specific treatment factors impacting outcomes in most psychological interventions [32]. One common explanation for treatment credibility and expectancy impacting treatment outcomes is that these increase individuals' engagement with interventions [33]. In the present study, the participants with higher levels of treatment expectancy and credibility might engage more with the cognitive tasks provided in the intervention and use the strategies they learned more frequently. However, it is also possible that believing in the beneficial effects of interventions increases placebo responses [34]. Nevertheless, it is important to ensure that interventions are presented with a logical coherent therapeutic framework and trustworthiness to ensure its credibility.

The hypothesis that demographic variables would not predict treatment response was not fully supported, as participants with

TABLE 3 | Results from linear mixed model analyses with estimated pre-treatment level and change in the BRIEF-A GEC adjusted for categorical variables.

Variable	Pre-treatment (95% CI)	*p*	Pre-treatment to FU (95% CI)	*p*
Male	116.80 (100.80–132.80)	0.037*	−10.12 (−27.89–7.65)	0.200
Female	18.12 (1.13–35.12)		−12.28 (−31.20–6.64)	
Partner	131.64 (124.73–138.54)	0.593	−28.18 (−35.52 to −20.84)	**0.001***
No partner	2.83 (−7.68–13.35)		19.71 (7.85–31.58)	
Higher education	131.94 (125.79–138.09)	0.512	−21.37 (−28.16 to −14.59)	0.710
No higher education	4.50 (−9.09–18.10)		−3.14 (−13.62–19.91)	
Depression ≤ 1 year	132.50 (12* 1.23–143.77)	0.942	−31.40 (−42.74 to −20.06)	0.031
Depression > 1 years	0.47 (−12.35–13.29)		14.73 (1.35–28.10)	

Note: Bold *p*-values are significant after Bonferroni correction.
Abbreviations: FU, six-month follow-up; Pre-treatment to FU, estimated change from pre-treatment assessment to follow-up on the BRIEF-A GEC and the difference in BRIEF-A GEC scores between predictor categories from pre-treatment to 6-month follow-up; Pre-treatment, the intercept level on the BRIEF-A GEC and the estimated difference on the BRIEF-A GEC between predictor categories at pre-treatment.
*$p \leq 0.05$.

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a partner reported larger changes in subjective executive functioning than those without a partner. Research has shown that social support is related to cognitive functioning, where significant others provide essential practical and emotional support in coping with cognitive deficits [35]. The results might therefore reflect the role of partners in encouraging participants to continue to apply the strategies learned during the intervention. This indicates that those participants lacking support from a partner are more vulnerable and might need more therapist guidance and follow-up in order to gain a positive treatment response. Partner status is also a known risk factor for relapse in depression [36], thus highlighting the need to be aware of this as a variable in interventions.

This study investigated predictors of treatment response to an internet-delivered cognitive-enhancement intervention for individuals in remission from depression. Similar predictors have been explored in a face-to-face format [20]. Despite the differing delivery methods, these studies identified illness duration as a predictor of treatment outcome. Treatment credibility and expectancy likewise influence response across internet-delivered and face-to-face mental-health interventions [31]. However, research shows that individuals receiving internet-delivered interventions report lower levels of treatment credibility and expectancy in the early course of treatment compared to face-to-face cognitive behavioral interventions, while this discrepancy diminishes as internet-delivered interventions progress [37]. It would therefore be of interest to examine how change in credibility and expectancy during treatment relates to treatment outcomes. Moreover, future research should investigate delivery format and level of therapist support, as these factors are of importance for increasing access to cognitive enhancement interventions in a cost-effective way.

4.1 | Strengths and Limitations

A strength of this study is that findings from two previous studies were replicated through explicitly stated hypotheses. However, there are also several limitations. The sample size was small, which has implications for the power and generalizability of the findings. Future research should focus on predictor studies with a larger sample to further examine the impact of variables that have been identified as significant in this and previous research. Moreover, by pooling the data from participants in both the intervention and control groups, we might have missed differences in predictors of treatment response between the groups. However, the primary aim of the study was to identify general predictors of treatment response across treated individuals. To this end, pooling the data increased the overall sample size and thereby improved statistical power. There are relevant variables that were not investigated in the current study. Inpatient status, substance abuse, and severity of depressive disorder should, for example, be considered when generalizing the results to other populations with a history of depression. The reliance on self-reports without complementing them with neuropsychological tests represents a limitation. The inclusion of neuropsychological tests as predictors or outcome measures is recommended to provide a more comprehensive understanding of treatment response.

5 | Conclusion

The finding that shorter duration of depression was associated with better outcomes underscores the importance of targeting subjective executive deficits early in the course of the depression trajectory. Furthermore, interventions should be developed to ensure credibility. Having a partner might be associated with improvement in subjective executive functioning, and the mechanisms should be investigated further. However, large-scale studies are still needed to identify predictors of treatment response in individuals remitted from depression undergoing cognitive enhancement interventions.

Author Contributions

Å.H., S.B.M., and T.N. were responsible for the conceptualization and design of the study. S.B.M. collected the data, conducted the statistical analyses with A.H., and wrote the main manuscript. E.H.R. contributed to data collection. All authors participated in the interpretation of data, manuscript review, and final approval.

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Ethics Statement

The study was approved by the Regional Committee for Medical Research Ethics of Western Norway (204287). Informed consent was provided by all participants. The study was conducted in accordance with the Declaration of Helsinki for Ethical Research [38].

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Peer Review

The peer review history for this article is available at https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/acps.70019.

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